



AI for Climate Adaptation Project Investment Case Creation





AI for Climate Adaptation Project Investment Case Creation



Co-funded by
the European Union

Climinvest Introduction
Richard McCreddie (University of Glasgow)

Agenda

- 09:30 AM - 09:45 AM
 - Welcome & Introduction to Green Investments and the CLIMINVEST Project
- 09:45 AM - 10:15 AM
 - The Bankability-by-Design Framework
- 10:15 AM - 10:35 AM
 - Overview of AI technologies within CLIMINVEST
- 10:35 AM - 10:55 AM
 - Information Extraction from Complex Documents
- 10:55 AM - 11:25 AM
 - Coffee break / Networking
- 11:25 AM - 11:45 AM
 - Financial Question Answering
- 11:45 AM - 12:05 AM
 - Bankability Report Generation
- 12:05 AM - 12:25 PM
 - Demonstration of the CLIMINVEST Studio
- 12:25 PM - 12:45 PM
 - The CLIMINVEST Learning Center & Marketplace
- 12:45 PM - 1:00 PM
 - Future directions and closing

What is Climinvest ?

Climinvest is a Horizon Europe project that aims to **reduce the barriers** to **funding green infrastructure projects**

- It's connected to the challenges that the European Commission encountered when implementing the **Green Deal**, i.e. it was proving difficult to get enough green infrastructure projects off the ground to meet their targets

The underlying theory of Climinvest is not that there is a lack of good ideas for green projects or people to implement them, but rather **too many projects fail to attract the needed funding to make them viable**

- There are many diverse reasons for this...
 - Lack of a proven business model
 - The right funder is not found
 - Projects fail to account for legal/governance compliance standards
 - Project co-benefits are poorly quantified
 - ... and so on

Can we fix this?

Investment Cases

The artifact at the centre of this is the **investment case**

- It's a long document (10-100 pages) that gets sent to your target funding agency

GCF DOCUMENTATION

PROJECTS

Funding Proposal

A PROJECT / PROGRAMME SUMMARY

GREEN CLIMATE FUND FUNDING PROPOSAL | PAGE 1 OF 45

A.1. Brief Project / Programme Information	
A.1.1. Project / programme title	Catalyzing private investment in sustainable energy in Argentina – Part 1
A.1.2. Project or programme	Project
A.1.3. Country (ies) / region	Argentina
A.1.4. National designated authority (ies)	National Directorate of Programs with International Credit Organizations (DNPOIC)
A.1.5. Accredited entity	Inter-American Development Bank
A.1.5.a. Access modality	☐ Direct ☒ International
A.1.6. Executing entity / beneficiary	Executing Entity: Inter-American Development Bank and Inter-American Investment Corporation Beneficiary: various renewable energy projects
A.1.7. Project size category (Total investment, million USD)	☐ Micro (<10) ☐ Small (10-50) ☐ Medium (50-250) ☐ Large (>250)
A.1.8. Mitigation / adaptation focus	☒ Mitigation ☐ Adaptation ☐ Cross-cutting
A.1.9. Date of submission	September 26 th , 2016
A.1.10. Project contact details	Contact person, position Gian Franco Carassale – Lead Investment Officer Alfredo Iadarte – Advisory Services Team Gloria Visconti – Climate Change Lead Specialist
	Organization Inter-American Development Bank
	Email address gianfranco@iadb.org aiadarte@iadb.org gloria@iadb.org
	Telephone number +1 202 623 3360
	Mailing address 1300 New York Ave NW, Washington DC 20577

A.1.11. Results areas (mark all that apply)	
Reduced emissions from:	
☒	Energy access and power generation (E.g. on-grid, on-grid or off-grid solar, wind, geothermal, etc.)
☐	Low emission transport (E.g. high-speed rail, rapid bus system, etc.)
☐	Buildings, cities and industries and appliances (E.g. new and retrofitted energy-efficient buildings, energy-efficient equipment for companies and supply chain management, etc.)
☐	Forestry and land use (E.g. forest conservation and management, agroforestry, agricultural viticulture, water treatment and management, etc.)
Increased resilience of:	
☐	Most vulnerable people and communities (E.g. mitigation of operational risk associated with climate change – diversification of supply sources and supply chain management, relocation of manufacturing facilities and warehouses, etc.)
☐	Health and well-being, and food and water security (E.g. climate-resilient crops, efficient irrigation systems, etc.)

B FINANCING / COST INFORMATION

GREEN CLIMATE FUND FUNDING PROPOSAL | PAGE 2 OF 45

B.1. Description of Financial Elements of the Project / Programme						
<i>Please provide a description of how the choice of financial instrument(s) will overcome barriers and achieve project objectives, and leverage public and/or private finance</i>						
The use of GCF investment resources will be aimed at helping structure and complete the long-term debt packages required for projects to be financially viable at competitive prices. For this, GCF resources will be made available in two modalities:						
1) long-term project debt , matching the tenor that the IDBG will provide.						
2) refinancing guarantees , which would help commercial lenders and commercial debt investors extend the tenor they can offer to those required by projects, otherwise generally restricted to 5-7 years. The guarantees would allow sponsors to accept financing that give these lenders the possibility to exercise early repayment options, by guaranteeing the refinancing of such loans. This instrument thus has the potential to crowd in commercial lenders and investors that would otherwise not be able to extend suitable terms and participate in the financing to deliver the volumes needed by the ambitious targets of the RenovAr public tendering program.						
Through these two GCF instruments, IDBG expects to help leverage additional, adequately termed debt and timely close projects, to reduce the incidence of drop outs (awarded projects that are never built) that has been so high in previous tenders (just about 15% of the awarded capacity built in the previous renewable energy tender, in 2010).						
The proposed USD 130M from the GCF would help mobilize investment for a total of about USD 650M, and support about 400 MW, which is equivalent to about 40% of the capacity originally targeted by the first tender under RenovAr. If achieved, the GCF Program will have helped provide a meaningful contribution to the success of the government initiative with long-term demonstration and capacity building effects. This would help pave the way for the significant replication and scale-up required in the mid and long term to achieve Law 27.191's ambitious renewable penetration goals.						
Component	Sub-component (if applicable)	Amount (for entire project)	Currency	Amount (for entire project)	Local currency	GCF funding amount
Component 1	Sub-component 1.1	650	million USD (\$)			130 million USD
	Financing of renewable energy power projects					USD
Component 2	Sub-component 2.1	1.0	million USD (\$)			1.0 million USD
	Enhancing policy, planning, regulation and project financing capacity					USD

E EXPECTED PERFORMANCE AGAINST INVESTMENT CRITERIA

GREEN CLIMATE FUND FUNDING PROPOSAL | PAGE 26 OF 45

<i>Please describe financial viability in the long run beyond the Fund intervention. Please describe the GCF's financial exit strategy in case of private sector operations (e.g. IPOs, trade sales, etc.).</i>	
The GCF support is required at this stage given the challenges associated to long-term investment in Argentina today (among other, still low – albeit improving – sovereign credit rating). In particular given the financial status of the energy sector. As the government continues to advance on various macroeconomic and sector reforms, these hurdles are expected to be overcome in the mid-term, resulting in a sector that can be financed on a fully commercial basis. Outstanding renewable energy resource conditions in Argentina, adequate availability of other required resources, and the continued decrease in technology cost should ensure financial viability of these investments once the sectoral and general investment challenges are overcome. A more detailed analysis on the pillars for long-term sustainability of GCF target investment and its exit strategy is provided in section D.2.	
E.6.4. Application of best practices	
<i>Please explain how best available technologies and practices are considered and applied. If applicable, specify the innovations/modifications/adjustments that are made based on industry best practices.</i>	
Based on IDBG's vast track-record financing wind and solar PV projects in Latin America, a number of best practices have been identified and consolidated by the IDBG credit team. These practices are considered by project teams and credit officers during due diligence and structuring of these transactions, and implemented as/when projects' specific conditions may allow. A number of them are already included in the credit and structuring requirements established for the Program, including those related to financial structuring (ratios, etc.), due diligence review, project contracts, warranties, etc.	
The other driver and incentive for use of best technologies (beyond any credit requirements IDBG will set as a lender) is the competitive nature of the process, where most efficient, cost effective and reliable technologies (considering the life cycle of plants) will have an advantage over competing options, first during the evaluation of technical proposals (where some may be discarded if not found to be compliant with minimum technical requirements), and secondly, and most importantly, during the evaluation and ranking of economic offers, as price will be the most determinant competitiveness factor of eligible offers and is closely linked with the levelized cost of energy of each project, which captures life cycle costs and performance.	
E.6.5. Key efficiency and effectiveness indicators	
Estimated cost per t CO ₂ e, defined as total investment cost / expected lifetime emission reductions (mitigation only)	
GCF core indicators	(a) Total project financing
	US\$ 650 Million
	(b) Requested GCF amount
	US\$ 130 Million (investment component)
	(c) Expected lifetime emission reductions overtime
US\$ 42.4 / tCO ₂ e	
(d) Estimated cost per tCO ₂ e (d = a / c)	
(e) Estimated GCF cost per tCO ₂ e removed (e = b / c)	
US\$ 8.5 / tCO ₂ e	
<i>Describe the detailed methodology used for calculating GHG emission reductions were calculated based on</i>	

in sustainable energy in Argentina -

IDB | Decision B.15/07

It's a lot to prepare...

² While this % is referred to the capacity tendered under the first RenovAr tender, GCF's support will likely me spread throughout the first few tenders, to optimize its value add.

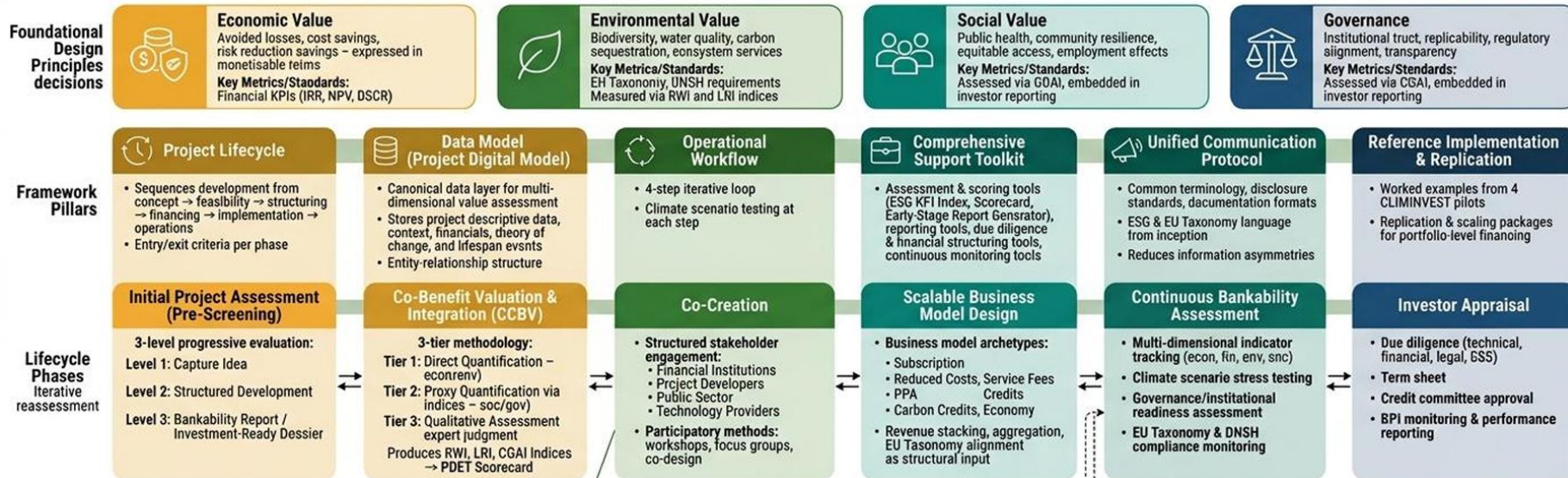
How can Climinvest help?



Project Intro

The first major contribution of Climinvest is the **Bankability-by-Design (BbD) framework**

- This is the theory part; it defines the **process** by which a person with a green project idea can develop that into a business case, and what we should **pay attention to** when doing so



How can Climinvest help?

However, Climinvest is not just a set of design documentation and recommendations, it aims to build the tools to make the Bankability-by-design process **as easy as possible**

- This is can never be as simple as 'AI please write me a good business case'
- ... and so the project is built around three main technological pillars:



Climinvest Academy

A learning platform with webinars, tutorials and courses designed to teach project developers and investors about Climinvest tooling, frameworks and best practices



Climinvest Marketplace

A hub for green investment projects from across the EU and beyond, providing rich information about their structure, activities, outcomes and impact, which can be replicated to new regions



Climinvest Assistant

An AI-powered platform to guide project developers from initial project concept, business-case development through investment case creation and beyond

Climinvest Scenarios



Let's imagine a scenario...

You work at a company that installs solar panels and you have a great idea for a green infrastructure project;

- You have been talking with farmers that maintain long irrigation networks
- If you could install solar panels over those networks, not only would you reduce wasted water from evaporation, but you also generate power that can be used to run nearby infrastructure!

So, now what do you do....



Let's imagine a scenario...

You tell the farmer about your idea, they are interested, but such an installation will have a large up-front-cost... and they don't have the capital

You need to find a new source of funding, and so you go to a local bank... and they think they might be able to raise capital for the venture, but they need you to put together an investment case

➤ They give you a template with some headings...

- Project overview and objective
- Project Promoter & Co-Creation Stakeholders
- Strategic Fit & Climate Rationale
- Technical Feasibility Assessment
- Financial Sustainability & Business Model
- ESG, Regulatory & Climate Compliance
- MRV & Impact Reporting Framework

The idea is probably feeling
less appealing by the minute

This is where good ideas fail, because the people that come up with the ideas don't have the knowledge to convert that to a viable business case that can attract funding

This is not a hypothetical scenario, Climinvest is working with SMEs across Europe to test whether the BbD Framework and developed tooling can help get more green projects off-the-ground



**Solar Overlay for
Water Adaptation,
Transformation and
Energy**

**Bankability
revamping and
repowering of large
scale Solar
Photovoltaic (PV)
assets**



**Predictive
bankability and
replicability of
wind energy projects
based on climatic
predictions
integration**

**Increasing the
Bankability of Old
Buildings
Retrofitting
Programs**



An aerial photograph of a large-scale solar farm. The solar panels are arranged in neat, parallel rows across a green field. The panels are tilted at an angle to maximize sunlight absorption. In the background, there are various agricultural fields, some with yellow crops like corn, and a few buildings under a clear sky. The overall scene depicts a sustainable energy project integrated into a rural landscape.

Climinvest Vision

This is where Climinvest comes in

Let's roll back a little, you needed funding, instead of directly going to a bank, Climinvest's bankability by design framework (more on this later) recommends that you first go through two light-weight processes



Project Pre-Screening

The goal of pre-screening is to perform an initial check that the project developer has thought enough about the project to provide some confidence that the project is feasible, covering basic information regarding the project aims, benefits and revenue model

If not...



Climinvest Academy

A learning platform with webinars, tutorials and courses designed to teach project developers and investors about Climinvest tooling, frameworks and best practices

The Academy can help them learn about what their project idea needs



Climinvest Marketplace

A hub for green investment projects from across the EU and beyond, providing rich information about their structure, activities, outcomes and impact, which can be replicated to new regions

The Marketplace provides project ideas and exemplars they can be inspired by

This is where Climinvest comes in

Let's roll back a little, you needed funding, instead of directly going to a bank, Climinvest's bankability by design framework (more on this later) recommends that you first go through two light-weight processes



Project Pre-Screening

The goal of pre-screening is to perform an initial check that the project developer has thought enough about the project to provide some confidence that the project is feasible, covering basic information regarding the project aims, benefits and revenue model

If yes



Funder Matching

Assuming the project looks viable, the next step is to see what funders are appropriate, its not just banks, maybe local government or a green investment fund might be more appropriate?

Project Concepting

Once the project owner has passed pre-screening and funder matching we can be fairly confident that the idea is feasible/fundable, we just need to put together a strong case for it

Remember all those scary technical sections? We need to guide the project developer through the process of collecting the right information for these

- We have Apps for that!
- Depending on the type of funder(s) that are suitable for the project, we can direct the project developer to specific apps to collect specialist information expected by those funders



Creating the Investment Case

Once we have (most) of the information we need to build an investment case, we can start writing!

Climinvest has a custom service to make this effective and easy; the Climinvest **Studio**

- This service will come with pre-built smart templates for a range of funders, designed to leverage the information collected so far to produce a complete and auditable investment case semi-automatically
- In this way, we can produce multiple investment cases for different funders from the same project digital model



Agenda

- 09:30 AM - 09:45 AM
 - Welcome & Introduction to Green Investments and the CLIMINVEST Project
- 09:45 AM - 10:15 AM
 - The Bankability-by-Design Framework
- 10:15 AM - 10:35 AM
 - Overview of AI technologies within CLIMINVEST
- 10:35 AM - 10:55 AM
 - Information Extraction from Complex Documents
- 10:55 AM - 11:25 AM
 - Coffee break / Networking
- 11:25 AM - 11:45 AM
 - Financial Question Answering
- 11:45 AM - 12:05 AM
 - Bankability Report Generation
- 12:05 AM - 12:25 PM
 - Demonstration of the CLIMINVEST Studio
- 12:25 PM - 12:45 PM
 - The CLIMINVEST Learning Center & Marketplace
- 12:45 PM - 1:00 PM
 - Future directions and closing

Agenda

- 09:30 AM - 09:45 AM
 - Welcome & Introduction to Green Investments and the CLIMINVEST Project
- 09:45 AM - 10:15 AM
 - The Bankability-by-Design Framework
- 10:15 AM - 10:35 AM
 - Overview of AI technologies within CLIMINVEST
- 10:35 AM - 10:55 AM
 - Information Extraction from Complex Documents
- 10:55 AM - 11:25 AM
 - Coffee break / Networking

- 11:25 AM - 11:45 AM
 - Financial Question Answering
- 11:45 AM - 12:05 AM
 - Bankability Report Generation



Climinvest **Designs**

MINVEST Studio

- 12:25 PM - 12:45 PM
 - The CLIMINVEST Learning Center & Marketplace
- 12:45 PM - 1:00 PM
 - Future directions and closing

Agenda

- 09:30 AM - 09:45 AM
 - Welcome & Introduction to Green CLIMINVEST Project
- 09:45 AM - 10:15 AM
 - CLIMINVEST Project framework
- 10:15 AM - 10:35 AM
 - Overview of AI technologies within CLIMINVEST
- 10:35 AM - 10:55 AM
 - Information Extraction from Complex Documents
- 10:55 AM - 11:25 AM
 - Coffee break / Networking



Climinvest

Tools

- 11:25 AM - 11:45 AM
 - Financial Question Answering
- 11:45 AM - 12:05 AM
 - Bankability Report Generation
- 12:05 PM - 12:25 PM
 - Demonstration of the CLIMINVEST Studio
- 12:25 PM - 12:45 PM
 - The CLIMINVEST Learning Center & Marketplace
- 12:45 PM - 1:00 PM
 - Future directions and closing

Agenda

- 09:30 AM - 09:45 AM
 - Welcome & Introduction to Green Investments and the CLIMINVEST Project
- 09:45 AM - 10:15 AM
 - The Bankability-by-Design Framework
- 10:15 AM - 10:35 AM
 - Overview of the CLIMINVEST Studio
- 10:35 AM - 10:55 AM
 - Information Extraction from Complex Documents
- 10:55 AM - 11:25 AM
 - Coffee break / Networking



Demonstrations

- 11:25 AM - 11:45 AM
 - Financial Question Answering
- 11:45 AM - 12:05 AM
 - Bankability Report Generation
- 12:05 AM - 12:25 PM
 - Demonstration of the CLIMINVEST Studio
- 12:25 PM - 12:45 PM
 - The CLIMINVEST Learning Center & Marketplace
- 12:45 PM - 1:00 PM
 - Future directions and closing